

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Green Commute Planner

Version 1.0

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Course: Software Engineering

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Green Commute Planner system. This document serves as a formal agreement between stakeholders and the development team and acts as a reference for system design, implementation, validation, and maintenance.

1.2 Scope

The Green Commute Planner is a web/mobile-based application designed to promote environmentally sustainable commuting. The system enables users to compare transportation modes based on distance, travel time, cost, and estimated carbon emissions and recommends the most eco-friendly option.

The scope of the project is limited to a Minimum Viable Product (MVP) to be completed within the academic semester timeline.

1.3 Definitions, Acronyms, and Abbreviations

CO₂ – Carbon Dioxide

API – Application Programming Interface

MVP – Minimum Viable Product

Eco-Score – Sustainability rating assigned to commute options

SRS – Software Requirements Specification

1.4 References

Statement of Work (SOW) – Green Commute Planner

IEEE 830 – Software Requirements Specification Standard

Software Engineering Course Guidelines

2. Overall Description

2.1 Product Perspective

The Green Commute Planner is a standalone application that integrates with external mapping services and a relational database system. The system calculates emissions using predefined emission factors and stores trip history for analysis.

2.2 Product Functions

The system shall:

- Provide user registration and authentication
- Accept source and destination inputs
- Generate multiple transportation options
- Calculate distance and travel time
- Calculate carbon emissions
- Compare transportation modes
- Recommend the most eco-friendly route
- Store trip history
- Provide dashboard analytics
- Allow administrators to manage emission factors

2.3 User Classes and Characteristics

General User

Users with basic familiarity with web or mobile applications who use the system for daily commute planning.

Administrator

Authorised personnel responsible for managing emission factors and system configuration.

2.4 Operating Environment

- Web browser (Chrome, Edge, Safari)
- Mobile App
- Backend application server
- Relational database management system
- Internet connectivity

2.5 Design and Implementation Constraints

- Project completion within academic semester timeline
- Limited development resources
- MVP-level implementation

2.6 Assumptions and Dependencies

- Mapping APIs are available and operational
- Users have internet access
- Emission factors are predefined and validated

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Account Management

FR-1: The system shall allow users to register using a valid email address and password.

FR-2: The system shall authenticate users using secure login credentials.

FR-3: The system shall allow users to reset their password.

3.1.2 Commute Planning

FR-4: The system shall allow users to enter source and destination locations.

FR-5: The system shall retrieve available transportation modes for the selected route.

FR-6: The system shall calculate estimated distance and travel time.

FR-7: The system shall calculate estimated carbon emissions using predefined emission factors.

FR-8: The system shall compare commute options based on carbon emissions.

FR-9: The system shall recommend the transportation option with the lowest carbon emission.

3.1.3 Trip Management

FR-10: The system shall store trip details in the database.

FR-11: The system shall allow users to view trip history.

FR-12: The system shall calculate cumulative carbon emissions saved.

FR-13: The system shall display dashboard analytics.

3.1.4 Administrative Functions

FR-14: The system shall allow administrators to update emission factors.

FR-15: The system shall allow administrators to manage transportation modes.

FR-16: The system shall allow administrators to view system usage statistics.

3.2 Non-Functional Requirements

3.2.1 Performance

NFR-1: The system shall respond within 3 seconds under normal operating conditions.

NFR-2: The system shall support at least 500 concurrent users.

3.2.2 Security

NFR-3: The system shall encrypt user passwords before storage.

NFR-4: The system shall restrict administrative access to authorized users only.

3.2.3 Usability

NFR-5: The system shall provide a user-friendly interface.

NFR-6: The system shall be mobile responsive.

3.2.4 Reliability

NFR-7: The system shall provide meaningful error messages for invalid inputs.

3.2.5 Maintainability

NFR-8: The system shall follow modular architecture principles.

4. Use Case Specifications

Use Case UC-01: Plan Commute

Primary Actor: User

Precondition: User is logged in

Postcondition: Recommended route is displayed

Main Flow:

1. User enters source location
2. User enters destination
3. System retrieves route options
4. System calculates emissions
5. System compares options
6. System displays recommended route

Alternate Flow:

If the location is invalid, the system displays an error message.

Use Case UC-02: Update Emission Factor

Primary Actor: Administrator

Precondition: Administrator is logged in

Postcondition: Emission factor updated successfully

Main Flow:

1. Administrator selects transport mode

2. Administrator updates emission value
3. System validates input
4. System updates database
5. Confirmation message displayed

5. System Models

(Insert diagrams here in this order, each on a new page)

Figure 1: Use Case Diagram

Figure 2: Class Diagram

Figure 3: Sequence Diagram – Plan Commute

Figure 4: Activity Diagram

Figure 5: DFD Level 0

Figure 6: DFD Level 1

Figure 7: Three-Tier System Architecture

Each figure must include a short description below it.

6. Data Requirements

6.1 User Entity

Attributes:

- user_id (Primary Key)
- name
- email
- password

6.2 Trip Entity

Attributes:

- trip_id (Primary Key)
- user_id (Foreign Key)
- source
- destination

- distance
- transport_mode
- emission

6.3 Transport Mode Entity

Attributes:

- mode_id
- mode_name
- emission_factor

7. Requirement Traceability Matrix

Requirement ID	Description	Module	Test Case ID
FR-1	User registration	Authentication	TC-01
FR-2	Login	Authentication	TC-02
FR-4	Enter route	Route Module	TC-03
FR-7	Emission calculation	Emission Module	TC-04
FR-9	Recommendation	Recommendation Engine	TC-05
FR-10	Store trip	Database	TC-06
FR-14	Update emission factor	Admin Module	TC-07
NFR-1	Performance requirement	System Performance	TC-08
NFR-3	Password encryption	Security Module	TC-09

8. Risk Analysis

Risk	Impact	Mitigation
API failure	Route generation unavailable	Use mock data
Timeline delay	Incomplete features	Prioritize core modules
Data inaccuracy	Incorrect emission results	Validate emission factors

9. Acceptance Criteria

The system shall be considered acceptable if:

- All functional requirements are implemented
- All non-functional requirements are satisfied
- All mapped test cases pass successfully
- The working prototype is delivered within the defined timeline

10. Future Enhancements

- Real-time traffic integration
- AI-based route optimization
- Gamification features
- Corporate carbon reporting dashboard

TEST CASE SPECIFICATION

Green Commute Planner

1. Test Case Summary Table

Test Case ID	Requirement ID	Test Case Title	Priority
TC-01	FR-1	User Registration	High
TC-02	FR-2	User Login	High
TC-03	FR-4	Enter Source & Destination	High
TC-04	FR-6	Distance & Time Calculation	High
TC-05	FR-7	Emission Calculation	High
TC-06	FR-9	Eco-Friendly Recommendation	High
TC-07	FR-10	Store Trip in Database	Medium
TC-08	FR-11	View Trip History	Medium
TC-09	FR-14	Update Emission Factor (Admin)	Medium
TC-10	NFR-1	Performance Test	High
TC-11	NFR-3	Password Encryption	High

2. Detailed Test Cases

TC-01: User Registration

Requirement ID: FR-1

Description: Verify that a new user can successfully register.

Field	Details
Preconditions	User is not registered
Test Steps	1. Open registration page 2. Enter valid email & password 3. Click Register
Expected Result	User account is created successfully
Postcondition	User record stored in database

Status	To be Tested
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TC-02: User Login

Requirement ID: FR-2

Field	Details
Preconditions	User account exists
Test Steps	1. Enter valid credentials 2. Click Login
Expected Result	User redirected to dashboard
Postcondition	Session created
Status	To be Tested

TC-03: Enter Source & Destination

Requirement ID: FR-4

Field	Details
Preconditions	User logged in
Test Steps	1. Enter valid source 2. Enter valid destination 3. Click Search
Expected Result	System generates route options
Postcondition	Route request processed
Status	To be Tested

TC-04: Distance & Time Calculation

Requirement ID: FR-6

Field	Details
Preconditions	Valid route exists
Test Steps	1. Request route 2. Check displayed distance & time
Expected Result	Correct distance and estimated time displayed
Status	To be Tested

TC-05: Emission Calculation

Requirement ID: FR-7

Field	Details
Preconditions	Route options available
Test Steps	1. Select transport mode 2. Verify emission value
Expected Result	Emission calculated correctly using emission factor
Status	To be Tested

TC-06: Eco-Friendly Recommendation

Requirement ID: FR-9

Field	Details
Preconditions	Multiple commute options available
Test Steps	1. Compare emissions 2. Check recommended route
Expected Result	System recommends lowest emission option
Status	To be Tested

TC-07: Store Trip

Requirement ID: FR-10

Field	Details
Preconditions	Trip completed
Test Steps	1. Plan commute 2. Verify database entry
Expected Result	Trip record saved successfully
Status	To be Tested

TC-08: View Trip History

Requirement ID: FR-11

Field	Details
Preconditions	User has previous trips
Test Steps	1. Open dashboard 2. View trip history

Expected Result	Correct trip records displayed
Status	To be Tested

TC-09: Update Emission Factor (Admin)

Requirement ID: FR-14

Field	Details
Preconditions	Admin logged in
Test Steps	1. Select transport mode 2. Modify emission factor 3. Save changes
Expected Result	Emission factor updated successfully
Status	To be Tested

TC-10: Performance Test

Requirement ID: NFR-1

Field	Details
Preconditions	System running under normal load
Test Steps	1. Request route 2. Measure response time
Expected Result	Response time < 3 seconds
Status	To be Tested

TC-11: Password Encryption

Requirement ID: NFR-3

Field	Details
Preconditions	User registers
Test Steps	1. Register new user 2. Check database password field
Expected Result	Password stored in encrypted format
Status	To be Tested